

Week 11

Synthesising Data and Reporting Results

- Learn to combine insights from multiple datasets into actionable findings
- Develop concise and effective data analysis reports
- Practice communicating trends and recommendations for diverse audiences

Learning Objectives

- Synthesise results from different data sources into coherent insights
- Write industry-standard data analysis and synthesis reports
- Communicate findings clearly to technical and non-technical stakeholders
- Prepare actionable recommendations based on analysis

Session Agenda

- Introduction to data synthesis and its industry importance
- Frameworks for combining insights from multiple sources
- Best practices in action-oriented reporting
- Visualising trends and constructing clear arguments
- Audience-focused communication
- Workshop; synthesising and reporting on sample data
- Assessment preparation; linking to upcoming report and project

What is Data Synthesis?

- Merging and reconciling information from varied sources for deeper insights
- Identifying common patterns, complementary findings, and contradictions
- Essential for complex projects in AI, machine learning, and business analytics

Why Synthesis Matters in Industry

- AI and large-scale analytics rely on integrating multiple datasets to train robust models
- Synthesis supports decision-making, accountability, and innovation
- Critical skill for roles in AI research labs, data engineering teams, and organisations using open data

Steps for Synthesising Data

1. Review findings from each dataset and analysis
2. Compare and contrast relationships, trends, and anomalies
3. Resolve inconsistencies and check for reliability
4. Combine insights to create a unified understanding
5. Identify implications and make recommendations

Example; Synthesising Insights Across Datasets

- Analyse a UCI dataset alongside open text or image data (e.g., COCO, LAION-5B)
- Compare trends from structured spreadsheet data and visual AI datasets
- Example; Synthesise sales data and customer feedback for a more complete business insight

Writing Clear and Concise Data Analysis Reports

- Use plain language; avoid jargon where possible
- Start with a summary of key findings and implications
- Highlight trends with visuals; charts, graphs, tables
- Support arguments with specific data points
- Recommend concrete next steps or actions

Audience-Focused Communication

- Tailor content for technical; Use precise terms and code outputs
- Tailor content for non-technical; Explain findings simply, focus on business or ethical impacts
- Use storytelling and context to engage diverse stakeholders

Visualising Trends and Supporting Your Story

- Use effective visuals; graphs, heatmaps, correlation plots
- Explain what each visual shows and why it matters
- Emphasise trends, outliers, and patterns important for decision-making

Industry Case Study

- Scenario; AI team integrates product usage logs, customer reviews, and external market data
- Synthesis leads to product improvement and more effective AI models
- Reporting delivers actionable insight to business and technical stakeholders

Practical Activity; Synthesis and Reporting Workshop

- Work in pairs or small groups
- Select two related open datasets
- Combine and compare findings; identify patterns or contradictions
- Draft a short report summarising synthesis and offering recommendations
- Share results for group feedback

Assessment Preparation; Research Report and Projects

- Synthesising multiple sources is key for your Week 10 research report and later projects
- Use clear structure; Introduction, Data/Methods, Findings, Synthesis, Recommendations
- Refer to lessons from Weeks 8 to 10; data documentation, EDA, and presentation
- Practice concise writing and visual reporting

Link to Previous Weeks

- Week 8; Documenting and versioning data workflows; supports reproducibility in synthesis
- Week 9; Compliance and ethics; ensure privacy and integrity when combining data
- Week 10; EDA; essential foundation for meaningful synthesis and actionable reporting

Key Takeaways

- Successful data analysis moves beyond single datasets; synthesis draws comprehensive insights
- Effective reports are concise, visual, and actionable
- Audience and context shape communication strategy
- Continuous practice with synthesis and reporting prepares you for workplace tasks and assessments

Next Steps

- Finish ongoing EDA and begin report synthesis for assessment
- Read recommended industry examples of data reports (linked in LMS)
- Prepare any questions for lecturer or hands-on support next week

Questions and Support

- What challenges do you face when combining datasets or writing reports?
- Bring draft ideas and data for feedback next session
- Support available in-class or by appointment; see LMS for details

